README — run_infer_experiment.py

Purpose

This script validates STAR-aligned BAM files, runs strandness inference using RSeQC's *infer_experiment.py*, and generates a summary table of inferred strandness for downstream RNA-seq analysis.

Pipeline Overview

1) Input Validation

Checks presence, size, and genome build annotation of the BED12 file.

Verifies that RSeQC's *infer_experiment.py* is available in PATH.

Finds all .bam files in the input directory.

Ensures each BAM: has a valid index (.bai) — creates one if missing; and is coordinate-sorted.

Only valid BAMs proceed to strandness inference.

2) Strandness Inference

Runs *infer_experiment.py* on each valid BAM using the provided BED12 file.

Saves output logs (*.infer.txt) in the results directory.

3) Result Summarization

Extracts the final strandness call from each .infer.txt file.

Writes a combined TSV summary with columns: Sample and Strandness.

Inputs

BAM Directory

`/home/chu25/dsmz/results/star`

STAR-aligned, coordinate-sorted BAM files.

BED12 Annotation File

`/home/chu25/data/hg38_bed12/hg38.refseq.bed12`

Must correspond to GRCh38/hg38.

Outputs

Strandness Inference Files (*.infer.txt)

`/home/chu25/dsmz/results/strandness`

Summary Table

`/home/chu25/dsmz/results/strandness/summary/strandness_summary.tsv`

Log File

```
/home/chu25/dsmz/logs/strandness/strandness_check.log
```

Contains validation results and any errors/warnings.

Example Summary (TSV)

```
Sample	Strandness
NG-13640	1++,1--,0+-,0-+
NG-29643	0++,0--,1+-,1-+
```

Notes

- The BED file must be for the same genome build as the BAM files.
- BAMs not coordinate-sorted are skipped.
- If strandedness is needed for quantification, this summary can guide alignment or feature counting options.
- This script does not perform alignment; it only checks and processes existing BAMs.